1.9.1 Tolerance

It is impossible to manufacture resistors to the exact values. For example, if we want to manufacture a resistor to the value of 100 ohms , after the production, the value may be a bit higher or lower. This is due to unavoidable manufacturing process. Hence, there is some lenience in production of resistors. It is referred as tolerance. Fourth colour code is the tolerance code. If it is gold, tolerance is 5%. If it is silver, tolerance is 10%.

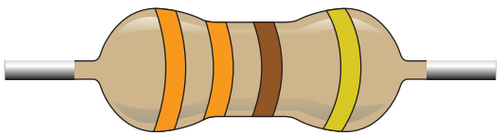


Figure 1.43 Tolerance code – 4th colour from left

Finding the values of resistors using colour codes

Example 1: Brown Black Brown



1 0 × 10^1
10 ×10
100 Ω

Example 2: Brown Red Black



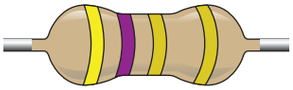
1 2 × 10^0
12 × 1
12 Ω

Example 3: Brown Black Red



1 0 × 10^2
10 ×100
1,000Ω = 1k Ω

Example 4: Yellow Violet Gold

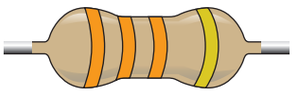


47×10^{-1}

$$\frac{47}{10}$$

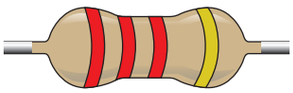
= 4.7 Ω

Example 5: Orange Orange Orange



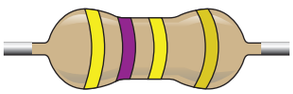
3 3 × 10^3
33,000Ω
33 kΩ

Example 6: Red Red Red



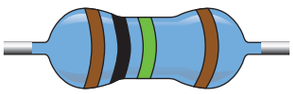
2 2 ×10^2
2200Ω
2.2kΩ

Example7: Yellow Violet Yellow



4 7 × 10^4
4,70,000Ω
470 kΩ

Example 8: Brown Black Green



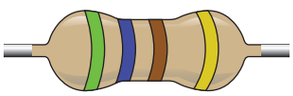
10 × 10^5
10,00,000Ω
= 1 MΩ

Example 9: Brown Black Black



1 0 × 10^0
10 × 1 = 10Ω

Example 10: Green Blue Brown



5 6× 10^1
560Ω

Figure 1.44 Finding the values of the resistors using colour coding

Tolerance Calculation

1. Find the tolerance value of the resistor given below

   Brown Black Brown Gold
   100Ω ± 5% = 100Ω ± 5Ω

Tolerance value of 100Ω = 95Ω to 105 Ω

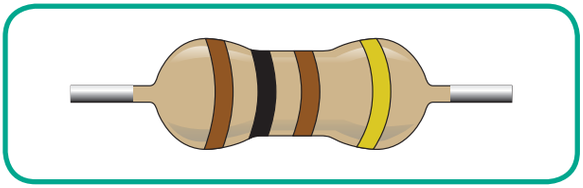


Figure 1.45(a) Calculating tolerance code

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